HARD · ADVANCED MATH

Nonlinear functions

04

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ADVANCED MATH

$$f(x) = (x - 1)(x + 3)(x - 2)$$

In the xy-plane, when the graph of the function f, where $y = f(x)$, is shifted up 6 units, the resulting graph is defined by the function g. If the graph of $y = g(x)$ crosses through the point $4, b$, where b is a constant, what is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN ADVANCED MATH

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function f, the table above shows several values of x and their corresponding values of

$$f(x)$$

, where a is a constant greater than 1. If k is a constant and

$$f(k) = a^{29}$$

, what is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

A right rectangular prism has a height of 9 inches. The length of the prism's base is x inches, which is 7 inches more than the width of the prism's base. Which function V gives the volume of the prism, in cubic inches, in terms of the length of the prism's base?

A. $Vx = xx + 9x + 7$

B. $Vx = xx + 9x$ - 7

C. $Vx = 9xx + 7$

D. $Vx = 9xx$ - 7

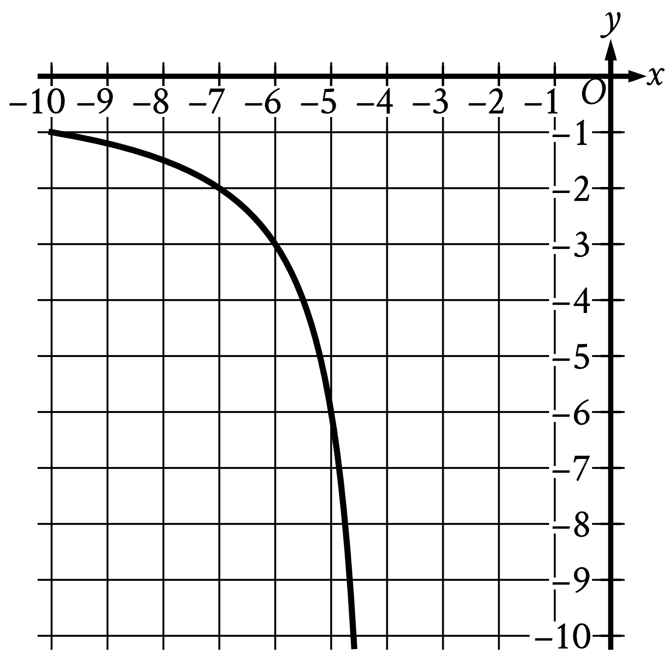
Q4 GRID-IN ADVANCED MATH

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = fx$ has an x-intercept at $2, 0$ and a y-intercept at $0, -323$. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right:
  - The curve is in quadrant 3.
  - The curve trends down sharply.
- As x increases, the curve approaches the line x equals negative 4.
- As x decreases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (negative 7 comma negative 2)
  - (negative 6 comma negative 3)
  - (negative 5 comma negative 6)

The rational function f is defined by an equation in the form $f x = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = f x$ is shown. If $g x = f x + 4$, which equation could define function g?

A. $g x = \frac{6}{x}$

B. $gx = \frac{6}{x+4}$

C. $gx = \frac{6}{x+8}$

D. $gx = \frac{6(x+4)}{x+4}$

Q6

ADVANCED MATH

$$fx = x - 10x + 13$$

The function f is defined by the given equation. For what value of x does fx reach its minimum?

A. -130

B. -13

C. $-\frac{23}{2}$

D. $-\frac{3}{2}$

Q7

ADVANCED MATH

$$fx = ax^2 + 4x + c$$

In the given quadratic function, a and c are constants. The graph of $y = fx$ in the xy-plane is a parabola that opens upward and has a vertex at the point h, k, where h and k are constants. If $k < 0$ and $f\text{-}9 = f3$, which of the following must be true?

I. $c < 0$
II. $a \geq 1$

A. I only

B. II only

C. I and II

D. Neither I nor II

Q8

GRID-IN ADVANCED MATH

The function f is defined by $fx = a^x + b$, where a and b are constants and $a > 0$. In the xy-plane, the graph of $y = fx$ has a y-intercept at $0, \text{-}25$ and passes through the point $2, 23$. What is the value of $a + b$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 GRID-IN ADVANCED MATH

A landscaper is designing a rectangular garden. The length of the garden is to be 5 feet longer than the width. If the area of the garden will be 104 square feet, what will be the length, in feet, of the garden?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

ADVANCED MATH

$$M = 1{,}800(1.02)^{t}$$

The equation above models the number of members, M, of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

A. $M = 1{,}800(1.02)^{\frac{q}{4}}$

B. $M = 1{,}800(1.02)^{4q}$

C. $M = 1{,}800(1.005)^{4q}$

D. $M = 1{,}800(1.082)^{q}$